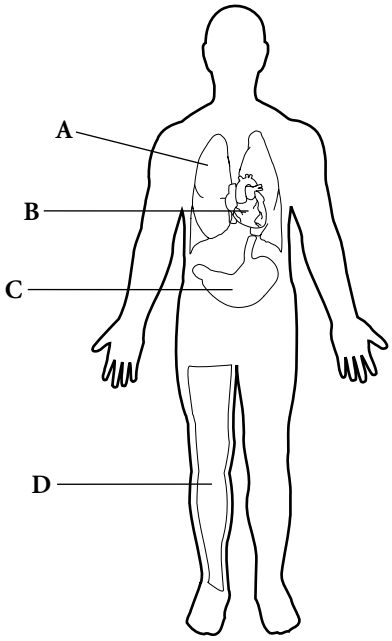
What are living things made from?

1 a Match the letter to the organs on the diagram. Choose from the list.



heart

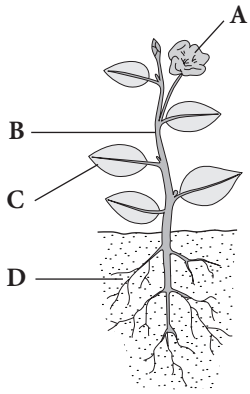
lung

leg

stomach

b For each organ labelled on the diagram write one sentence describing what it does.

2 a Match the letter to the organ:



stem

leaf

flower

roots

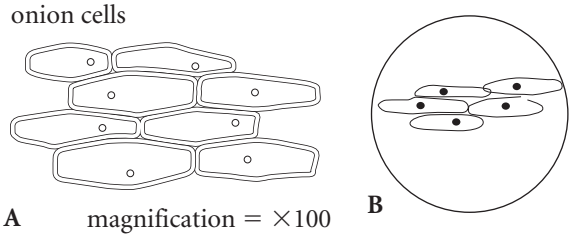
b For each organ labelled on the diagram write one sentence describing what it does.

3 Organs are made of lots of tiny building blocks. What is the name of these building blocks?

4 Where would you find tissues and what are they made of?

Using the microscope

Look at the two drawings of an onion cell. **A** is drawn properly. **B** is not.



1 By comparing the two sets of drawings, explain what you must do to draw microscope specimens properly.

2 What do the following parts of the microscope do?

a eyepiece and objective lenses

b stage clips **c** focusing knob

3 a John looks at a specimen which appears 100 times bigger than it actually is. What magnification is he using?

b Muna is looking at a leaf under her microscope. She is using an objective lens with magnification ×40 and an eyepiece lens with magnification ×10. What is the total magnification?

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What are cells like?

1 Plant and animal cells look different.

a Copy the table. Put a tick or a cross in the first two columns of the table to show which type of cell has which characteristic.

b In the final column of the table, write a short sentence to describe the function of each part of the cell.

	plant cell	animal cell	function
nucleus			
chloroplast			
cell membrane			
cytoplasm			
cell wall			

2 Read the passage and answer the questions which follow:

In the seventeenth century, Robert Hooke became the first person to see the world through a microscope. In his book *Micrographia*, Hooke describes how pieces of cork are actually made of tiny boxes which he called *cells*. As time went on, techniques for preserving tissue were improved, tools for slicing tissue thinly were invented, and lenses got much much better. This helped scientists see lots of detail under the microscope. In the nineteenth century, Matthias Jakob Schleiden and Theodor Schwann examined hundreds of different animals and plants and concluded that they were all made of cells. Schwann also realised that each cell could divide to make two new cells (people had believed that new cells just appeared ''as if by magic''!). This showed that it was not just the whole animal that was alive, but each cell was living too! It took Rudolph Virchow in 1858 to state the cell theory clearly:

Every animal (and plant) appears as a sum of its vital units, each of which bears in itself the complete characteristics of life.

a Who was the first person to use a microscope?

b Why does a specimen need to be thin to see it under the microscope?

c Why does a microscope have lenses?

d Cells can reproduce to make new cells. Who discovered this?

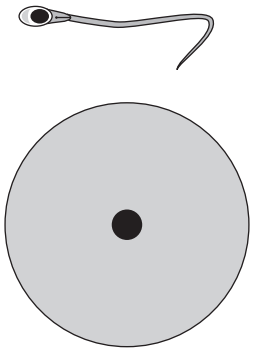
e What are the ''vital units'' mentioned by Virchow in his cell theory?

f Before the cell theory was introduced, people were *vitalists*. This means they thought that no single part of an animal or plant was alive, and that animals and plants were not made of cells. Write an article for the science page of a newspaper in 1859. In the article, you must explain what has been discovered about cells, and try to persuade the vitalists to change their beliefs.

ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

What do cells do?

All cells do not look the same. Some cells are specialised to do particular jobs. This means they may have special features which help them do their job properly.

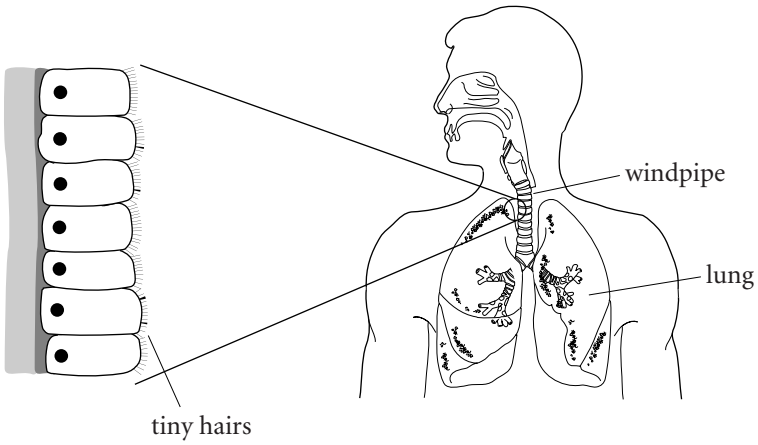


1 A young fish is made when an egg cell and a sperm cell join together in the water. The sperm cell has to swim through the water to find the egg. If the egg and sperm join together and a new fish starts to develop, the egg needs to provide food for it to grow until it hatches. Look at the diagram of the egg and sperm cells. For each one describe how it is specialised to do its job.

2 The nerve cells which carry messages around our body are very long and are branched at each end. Explain why.

3 Red blood cells carry oxygen around the body. It is important that each one carries as much oxygen as possible. By thinking about this, explain why red blood cells have no nucleus?

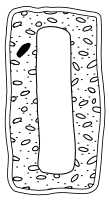
4 Epithelium cells are usually flat and cover the surface of organs.
There is a special type of epithelial cell which is shown in the diagram.
Millions of these cells line our windpipe and help stop dust travelling all the way into our lungs.
Once the dust has been stopped they also push it back up the windpipe to the throat. How do you think they do this?



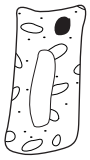
ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

Tissues and organs

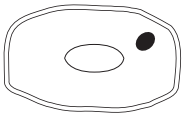
In humans, lots of nerve cells make nerve tissue and lots of muscle cells make muscle tissue. There are also tissues in plants. Look at the diagrams showing the different cells which you can find inside a leaf:



Palisade cell

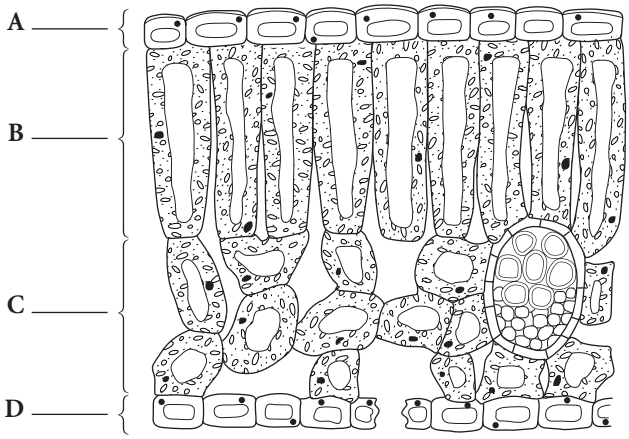


Spongy mesophyll cell



Epidermal cell

1 Use these diagrams to help you write down the name of each tissue (**A**, **B**, **C**, **D**) in the cross-section of the leaf. A cross-section lets us look inside the leaf at its tissues.



2 Copy and complete the passage using the given words:

protects	**transparent**	**upper epidermis**	**gases**	**spongy**
palisade	**light**	**epidermis**	**tissue**	

A leaf has several layers of _____. The _____ is the tissue at the top of the leaf. Its cells are thin and _____. The palisade tissue is made of _____ cells. Their job is to catch _____ energy to make food. Below the palisade tissue is the _____ mesophyll tissue. It can also make food, but its cells have gaps in between to let _____ in and out of the leaf. The lower _____ is at the bottom of the leaf. It holds the leaf together and _____ it.

3 Write down one function of the upper and lower epidermis.

4 Which part of a plant cell catches light energy to make food?

5 Which two types of tissue can make food?

6 Why is it important that the upper epidermis cells are thin and transparent?

7 Why is it important that the palisade tissue is near the top of the leaf?

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How are new cells made?

1 **a** Copy out these sentences in the correct order to show what happens during cell division. Next to each sentence, draw a diagram to show what happens at each stage.

- The cell starts to divide into two.
- The nucleus divides first.
- The cell finishes dividing. Two cells have been produced.
- There are two copies of the nucleus in the cell.

b Why does the nucleus divide first during cell division?

2 Copy and complete this passage using the words below:

pollination	**sex cells**	**egg**	**cells**	**fertilisation**

_____ is the transfer of pollen grains from one flower to another. Pollen grains carry pollen _____. These are the male _____ or gametes. When the nucleus of a pollen cell joins together with the nucleus of a female _____ cell, _____ happens.

3 Copy out these sentences in the correct order to show what happens between pollination and fertilisation:

- Pollen grain lands on the stigma of a flower.
- Pollen cell nucleus joins together with the egg cell nucleus.
- Pollen tube grows down into the ovary.
- Pollen tube grows from the pollen grain into the stigma.
- Pollen cell moves down the pollen tube to the ovary.

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